Ian Sendelbach

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EDUCATION

The University of Texas at Austin — B.S. Electrical and Computer Engineering **Expected Spring 2029**

University of Oklahoma — Engineering coursework prior to transfer; GPA: 4.00 **Aug 2025 – May 2026**

Relevant Coursework: Intro to EE (ECE 302), Intro to Computing (ECE 306), Digital Logic Design, ML for Power Systems

RESEARCH EXPERIENCE

Undergraduate Researcher, School of ECE, University of Oklahoma **Jan 2026 – May 2026**

- **First-author paper accepted at IEEE TPEC 2027** (presenting Feb 2027): "A Random Forest Approach to Voltage Deviation Management in Electric Vehicle Integrated Distribution Feeders," funded by an undergraduate research grant.
- Built a Python/MATPOWER batch simulation injecting randomized EV loads across thousands of IEEE 14 and 33-bus grid scenarios; trained a 30-tree Random Forest predicting voltage drop with R^2 = 0.977 and 0.0017 p.u. MAE.
- Ported the model to a TI C2000 F28P55x MCU as static C++ (EloquentML), using 65 KB of flash (25%) with 177 µs inference latency, leaving headroom for real-time PWM and ADC tasks.
- Designed a predictive closed-loop PWM throttling controller and processor-in-the-loop (PIL) testbed; under a sudden EV fleet load, it held voltage at the 0.95 p.u. ANSI C84.1 limit while the unmitigated grid collapsed below 0.65 p.u.

ENGINEERING EXPERIENCE

Controls & Embedded Engineer, Texas Guadaloop Levitation Team, UT Austin **Aug 2026 – Present**

- Built a modular Python/MuJoCo digital twin of a 16-state, 4-coil maglev pod with hardware-realistic motor driver, inductive gap, current, and Hall-effect sensor models, so the controller sees only firmware-level signals.
- Designed a cascaded control architecture (PID outer loop, Hall-feedback force loop, 4.8 kHz PI current loop), cutting post-disturbance gap ringing from 1.4 mm to 0.03 mm and peak deviation by 40% in simulation.
- Validated a 6th-order polynomial force model against 4,471 Ansys FEA points (0.19 N RMSE) and corrected mass/equilibrium modeling, eliminating 4.35 A/coil of holding current and widening tolerated force-model error from −7% to −12%.
- Implemented a parametric step-target setpoint generator and robustness tests (parameter perturbation, coil heating, supply sag) to support sim2real calibration against rig data.

Vision Team Lead, IEEE Robotics Team, University of Oklahoma **Aug 2025 – Jun 2026**

- Configured a Raspberry Pi 5 Linux vision coprocessor, resolving libcamera/PhotonVision conflicts for reliable headless boot.
- Interfaced the Pi with Arduino microcontrollers over I2C for autonomous real-time target tracking, and wrote firmware for analog Hall-effect sensors achieving 100% sorting accuracy across bench trials.

Hacklahoma 2026 — 2nd Place, Hardware Track **Feb 2026**

- Engineered an ESP32-C6 wearable for hikers integrating GPS and an OLED display over I2C for live speed and predicted-energy feedback; trained the energy model in scikit-learn on 100,000 synthetic hikes and led ML-to-embedded integration in 24 hours.

Small Team Captain, Sooner Competitive Robotics **Aug 2025 – Jun 2026**

- Led design of a <150 g combat robot in Fusion 360, with tolerance analysis for PLA shrinkage to fit the PCB and drivetrain.

PROJECTS

Article Quizzer — Chrome Extension (published, monetized) **Summer 2025**

- Built and published an OpenAI-powered quiz-generating Chrome extension; owned support and billing, peaking at ~$500 MRR.

LEADERSHIP

Projects Assistant Officer, IEEE Student Branch, UT Austin **Aug 2026 – Present**

- Designed and led technical workshops on MATLAB and control theory, each attended by ~30 students.

Event Committee Chair, IEEE Student Branch, University of Oklahoma **Sept 2025 – Jun 2026**

- Designed and led PCB soldering and embedded C workshops for ~30 students.

SKILLS

Controls & Modeling: PID, cascaded current/force loops, linearization, state-space modeling, digital twins, sim2real, MuJoCo

Embedded & Hardware: TI C2000 (F28P55x), Code Composer Studio, ESP32, Arduino, Raspberry Pi, I2C, UART, oscilloscope

Languages & OS: C, Python, MATLAB, JavaScript, Kotlin, Bash, Linux, Git

ML & Simulation: scikit-learn, Random Forest, EloquentML, MATPOWER, processor-in-the-loop (PIL) validation

CAD & EDA: KiCad, Fusion 360, Autodesk Inventor, Multisim, Ansys